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COURSE NAME – Operation System
COURSE CODE – CSA0408

Lab Observations

EXPERIMENT - 21

Develop a C program to implement worst fit algorithm of memory management.

PROGRAM:

```
#include <stdio.h>

#define MAX_MEMORY 1000

int memory[MAX_MEMORY];

void initializeMemory()
{
    for (int i = 0; i < MAX_MEMORY; i++)
        memory[i] = -1;
}

void displayMemory()
{
    int i = 0;

    printf("\nMemory Status:\n");

    while (i < MAX_MEMORY)
    {
        int start = i;
        int value = memory[i];

        while (i < MAX_MEMORY && memory[i] == value)
            i++;

        if (value == -1)
```

```
            printf("Free block: %d-%d\n", start, i - 1);
        else
            printf("Process %d: %d-%d\n", value, start, i - 1);
    }
}

void allocateMemory(int processId, int size)
{
    int bestStart = -1, bestSize = 0;
    int i = 0;

    while (i < MAX_MEMORY)
    {
        if (memory[i] == -1)
        {
            int start = i;

            while (i < MAX_MEMORY && memory[i] == -1)
                i++;

            int blockSize = i - start;

            if (blockSize >= size && blockSize > bestSize)
            {
                bestSize = blockSize;
                bestStart = start;
            }
        }
        else
            i++;
    }

    if (bestStart == -1)
    {
        printf("Allocation failed for Process %d\n", processId);
        return;
    }

    for (i = bestStart; i < bestStart + size; i++)
        memory[i] = processId;

    printf("Allocated %d units to Process %d at %d-%d\n",
           size, processId, bestStart, bestStart + size - 1);
```

```
}

void deallocateMemory(int processId)
{
    for (int i = 0; i < MAX_MEMORY; i++)
    {
        if (memory[i] == processId)
            memory[i] = -1;
    }

    printf("Memory released by Process %d\n", processId);
}

int main()
{
    initializeMemory();

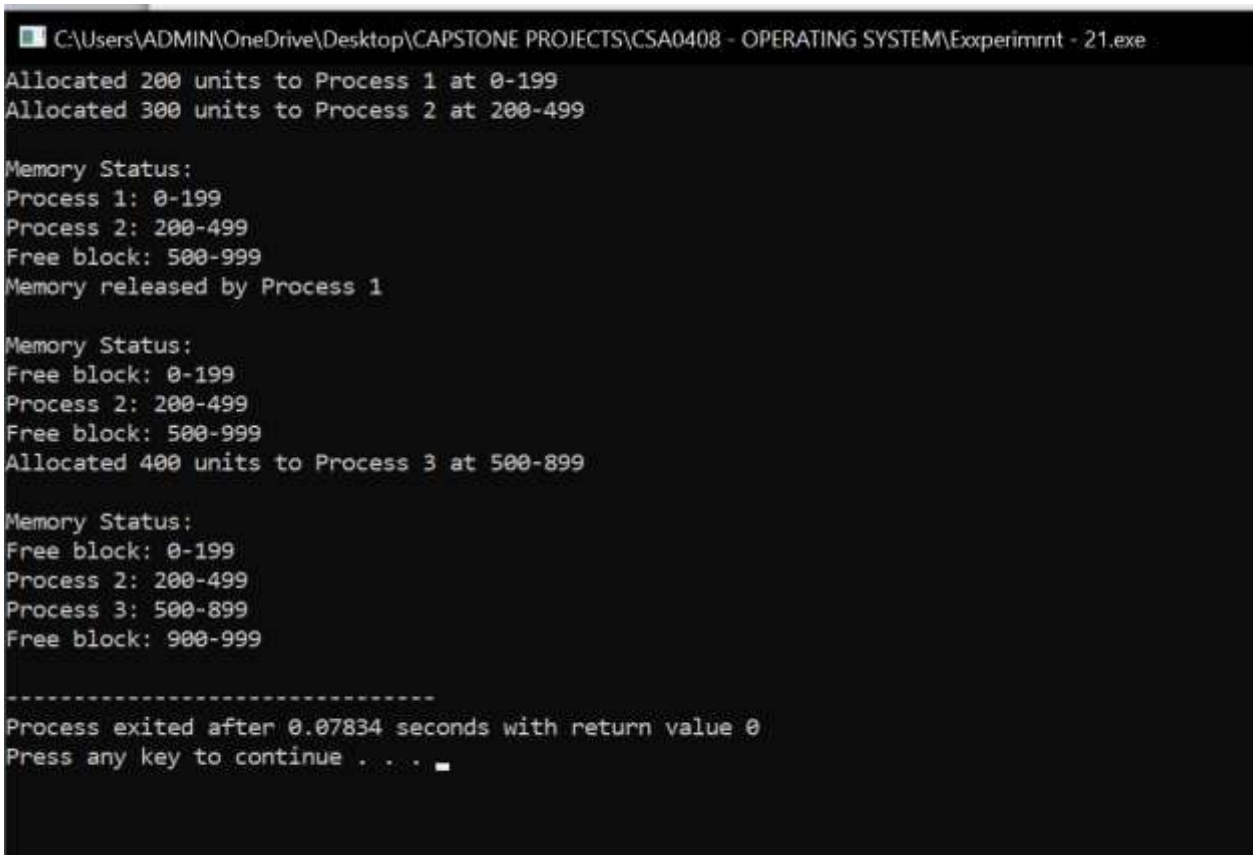
    allocateMemory(1, 200);
    allocateMemory(2, 300);
    displayMemory();

    deallocateMemory(1);
    displayMemory();

    allocateMemory(3, 400);
    displayMemory();

    return 0;
}
```

OUTPUT:


C:\Users\ADMIN\OneDrive\Desktop\CAPSTONE PROJECTS\CSA0408 - OPERATING SYSTEM\Exxperimrnt - 21.exe
Allocated 200 units to Process 1 at 0-199
Allocated 300 units to Process 2 at 200-499
Memory Status:
Process 1: 0-199
Process 2: 200-499
Free block: 500-999
Memory released by Process 1
Memory Status:
Free block: 0-199
Process 2: 200-499
Free block: 500-999
Allocated 400 units to Process 3 at 500-899
Memory Status:
Free block: 0-199
Process 2: 200-499
Process 3: 500-899
Free block: 900-999

Process exited after 0.07834 seconds with return value 0
Press any key to continue . . .